

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eighth Semester of B. Tech (ME) Examination

MAY 2016

ME421 Automobile Engineering

Date: 07.06.2017, Wenesday

Time: 01.30 p.m. To 04.30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the following short questions below. [07]

- a. What is an automobile?
- b. Give examples of Indian manufacturers of passenger cars.
- c. Enlist various types of frames.
- d. Why is frame narrowed at the front?
- e. What is meant by “wheel base”?
- f. What is wet clutch?
- g. Why sliding mesh gear box is not preferred?

Q-2.(a) Explain briefly the main forces which oppose the motion of a vehicle. [04]

(b) Give the working of a multi plate clutch with neat sketch? [05]

(c) Explain with neat sketch sliding mesh gear box. [05]

Q-3.(a) What do you understand by automobile engineering? Give brief history of automobile. [04]

Q-3.(b) Answer any two. [10]

- i. What are functions and requirements of a good tyre.
- ii. What are battery troubles and their remedies?
- iii. Write a short note on emission standards.

SECTION – II

Q - 4 Answer the question below. [07]

- a. What is type of suspension used in heavy duty vehicles?
- b. If a gear box is noisy what items should be checked?
- c. What is a Hotchkiss drive?
- d. What is a universal joint?

- e. What is a “differential”?
- f. What is Transaxle?
- g. What is collapsible steering column?

Q-5.(a) What are the functions of the suspension system? **[04]**

(b) Give classification of brakes. Explain with neat sketch hydraulic braking system. **[05]**

(c) Explain briefly with neat sketches. Any two type of following rear axles : **[05]**

(i) Half floating axle. (ii) Fully floating axle (iii) Three quarter floating axle

Q-6.(a) Define the following **[04]**

(i) Camber angle (ii) Caster angle (iii) King-pin inclination (iv) Toe-in

Q-6 (b) Answer any two. **[10]**

- i. Describe working of hydraulic torque converter.
- ii. Explain briefly construction of torque tube type of propeller shaft.
- iii. Is preventive maintenance cheaper? Explain.
